WORKSHEET

Chapter 7: Applications of simple interest

Round answers to two decimal places where appropriate.

1 Rebecca invests $800 000 at 7.5% p.a. simple interest for 6 years. How much interest will she earn?

2 Serina invests $150 for 18 months at a simple interest rate of 8.25% p.a. How much interest does she earn?

3 Jeremy invests $25 000 at a simple interest rate of 11.2% p.a. What will be the total value of his investment after 5 months?

4 Kimina invests $8000 for 6 years at 5% p.a. simple interest.

a How much interest will she earn each year?

b How much interest will she earn over the 6 years?

c What will be the total value of her investment at the end of 6 years?

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5 Marcus is able to make an investment at a high rate of 12% p.a. simple interest. How much does he need to invest to earn $15 000 interest in 3 years?

6 Lucy has $2000 to invest. She puts it into an account earning 3.5% simple interest per annum. How long should it be invested in order to earn $500 interest? Round up to the nearest year.

7 At what interest rate should $90 000 be invested if it is to earn $15 000 simple interest in 5 years? Answer correct to two decimal places.

8 Kevin is a multi-millionaire. If he invests $7 000 000 at 9% p.a. simple interest, how much interest will he earn per day?

9 Which of the following investments will yield the most interest?

A $8800 invested at 3% p.a. simple interest for 10 and a half years.

B $30 000 invested for 2 years and 1 month at 5% p.a. simple interest.

C $18 000 invested at 2.5% p.a. simple interest for 7 years.

10 Ari has $10 000 to invest. He is considering two options:

A Westpoint Bank offers a 6-month short-term deposit account at 1% simple interest per month with no account-keeping fees.

B Northgate Bank offers a 6-month short-term deposit earning 1.2% simple interest per month, but charges $10 per month in account-keeping fees.

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a What will be the value of Ari's investment after 6 months if he chooses to invest with Westpoint Bank?

b Which Bank should Ari invest with in order to maximise the value of his investment? How much more interest will he earn if he chooses this investment option?

11 Kyah borrows $5000 to buy a car. He is charged 8% simple interest per annum. He repays the loan in equal monthly instalments over 3 years.

a How much interest does Kyah pay?

b How much does Kyah pay in total for his car?

c Calculate the amount of his monthly instalment.

12 Percy and Annabeth borrowed $25 000 to pay for their wedding. They were charged 14% simple interest per annum, and repaid the loan in equal monthly instalments over 5 years.

a How much interest did Percy and Annabeth pay for their wedding?

b What was the total cost of their wedding?

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c How much were their monthly instalments?

13 Eva has a credit card that charges simple interest of 18.4% p.a. and has no interest-free period. She bought a laptop for $955 on 5 April and paid her account in full on 12 May. Interest is charged daily from the date of purchase to the date of payment inclusive.

a How many days did she pay interest on the purchase?

b Calculate the total interest charged.

c Find the total amount she paid for the laptop.

14 Mohammad bought a mobile phone on 23 July for $550 using his credit card. He is charged 19.8% p.a. simple interest on purchases that are not repaid within 7 days (inclusive of the purchase date). After the interest-free period, interest is calculated daily. He paid the account on 10 August.

a On which date did Mohammad begin to pay interest on his purchase?

b For how many days did he pay interest on the purchase?

c Calculate the total interest charged.

d Find the total amount he paid for his phone.

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Answers

1 $360 000

2 $18.56

3 $26 166.67

4 **a** $400

b $2400

c $10 400

5 $41 666.67

6 8 years

7 3.33% p.a.

8 $1726.03

9 C

10 **a** $10 600

b Northgate, $60

11 **a** $1200

b $6200

c $172.22

12 **a** $17 500

b $42 500

c $708.33

13 **a** 38 days

b $18.29

c $973.29

14 **a** 30 July

b 12

c $3.58

d $553.58

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